INCH-POUND
MIL-PRF-21480/12F(AS)
6 November 2012
SUPERSEDING
MIL-G-21480/12E(AS)
15 July 1983

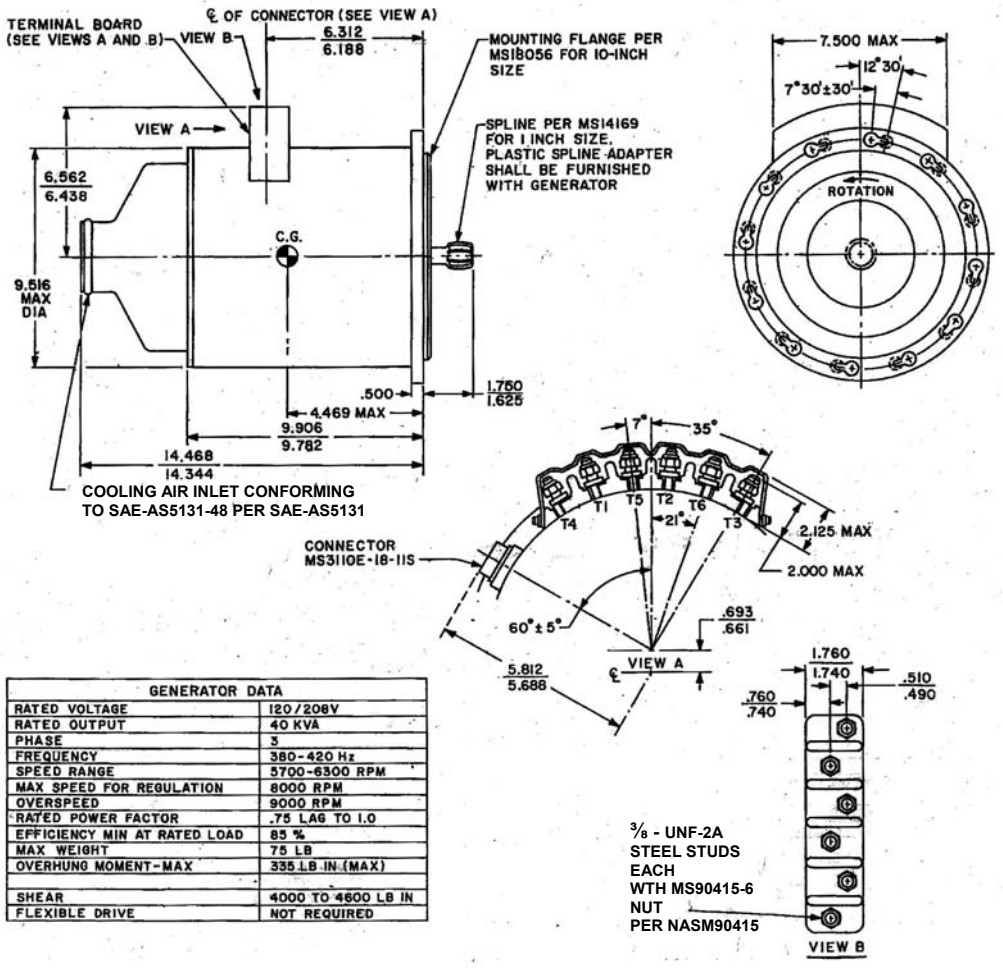
PERFORMANCE SPECIFICATION SHEET

GENERATOR SYSTEM, 40 KVA, 400 HZ AC INTEGRALLY EXCITED, BRUSHLESS TYPE

Inactive for new design after 21 August 1996.

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

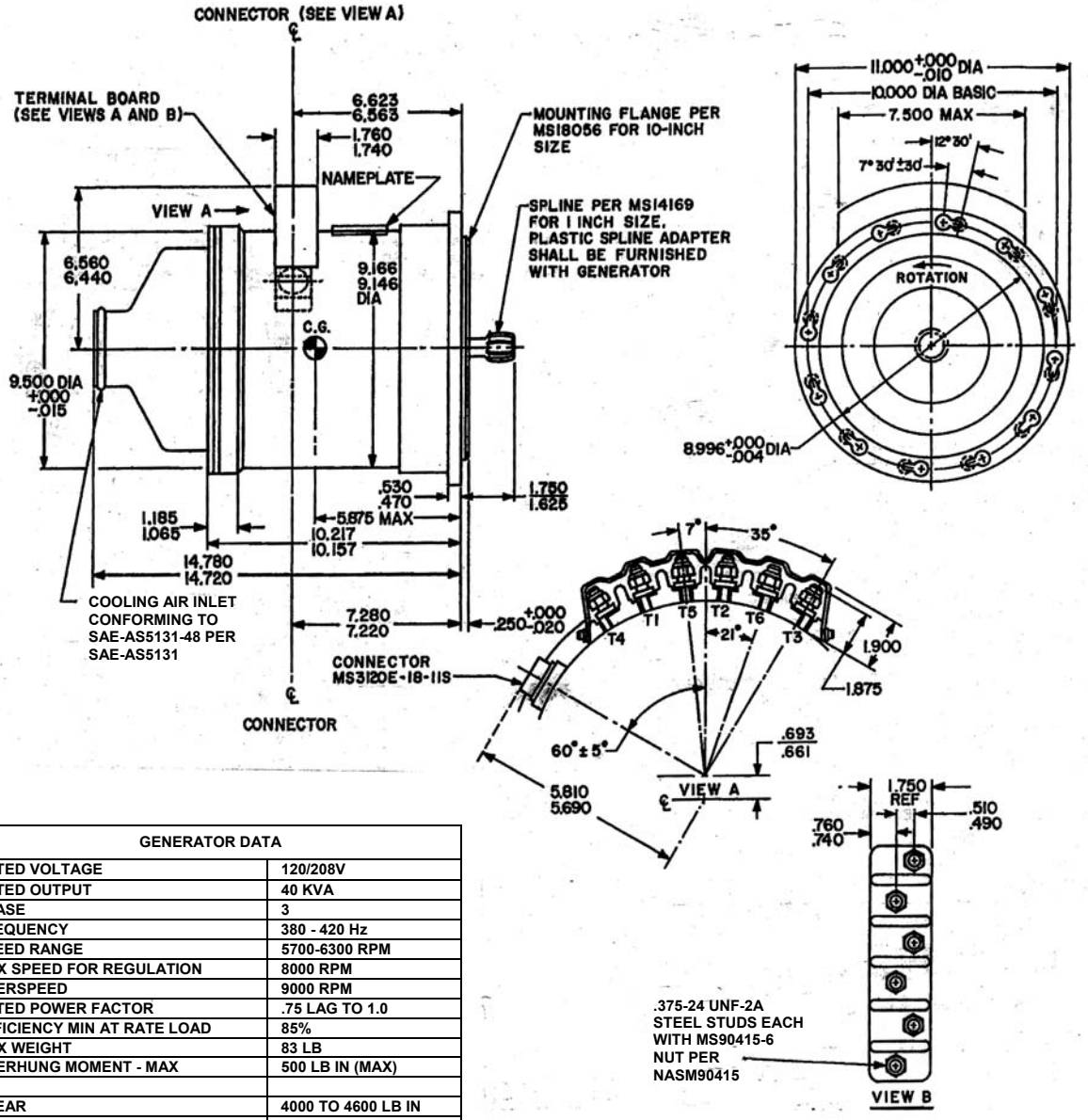
The complete requirements for procuring the generator described herein shall consist of this document and MIL-PRF-21480.



GENERATOR DATA	
RATED VOLTAGE	120/208V
RATED OUTPUT	40 KVA
PHASE	3
FREQUENCY	380-420 Hz
SPEED RANGE	5700-6300 RPM
MAX SPEED FOR REGULATION	8000 RPM
OVERSPEED	9000 RPM
RATED POWER FACTOR	.75 LAG TO 1.0
EFFICIENCY MIN AT RATED LOAD	85 %
MAX WEIGHT	75 LB
OVERHUNG MOMENT-MAX	335 LB IN (MAX)
SHEAR	4000 TO 4600 LB IN
FLEXIBLE DRIVE	NOT REQUIRED

FIGURE 1. M21480/12-1 generator.

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GENERATOR DATA	
RATED VOLTAGE	120/208V
RATED OUTPUT	40 KVA
PHASE	3
FREQUENCY	380 - 420 Hz
SPEED RANGE	5700-6300 RPM
MAX SPEED FOR REGULATION	8000 RPM
OVERSPEED	9000 RPM
RATED POWER FACTOR	.75 LAG TO 1.0
EFFICIENCY MIN AT RATE LOAD	85%
MAX WEIGHT	83 LB
OVERHUNG MOMENT - MAX	500 LB IN (MAX)
SHEAR	4000 TO 4600 LB IN
FLEXIBLE DRIVE	NOT REQUIRED

FIGURE 2. M21480/12-2, generator.

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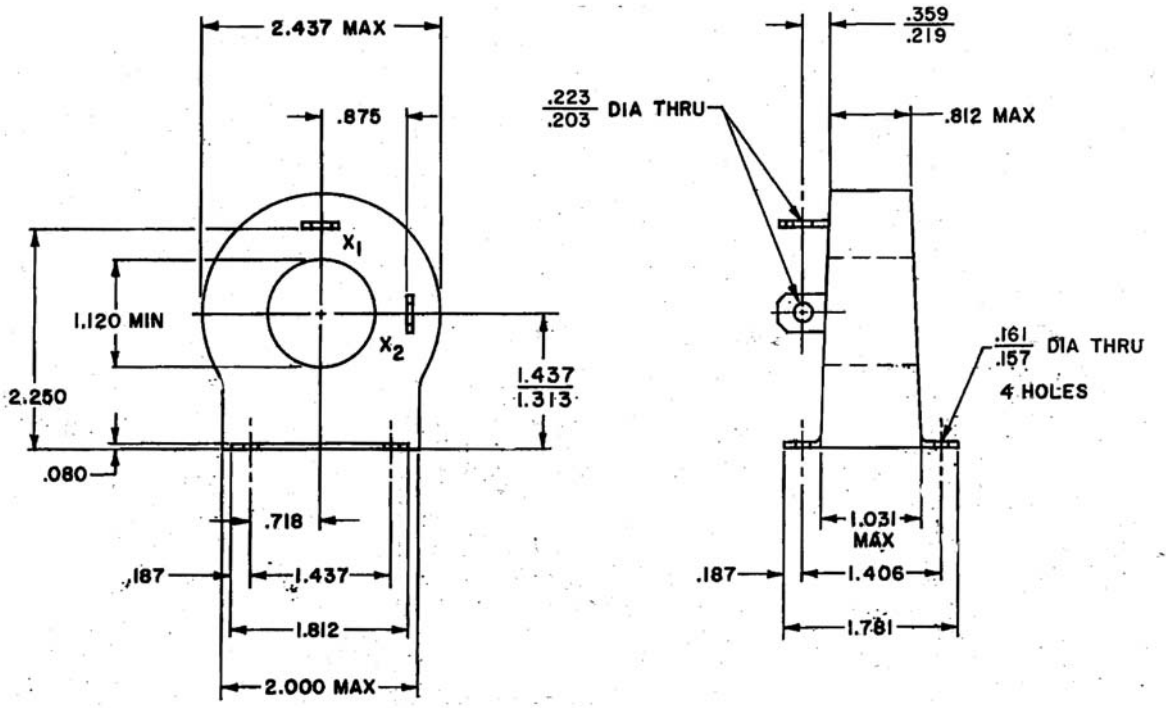


FIGURE 3. Load end current transformer, weight 0.4 pound max, P/N M21480/12-5.

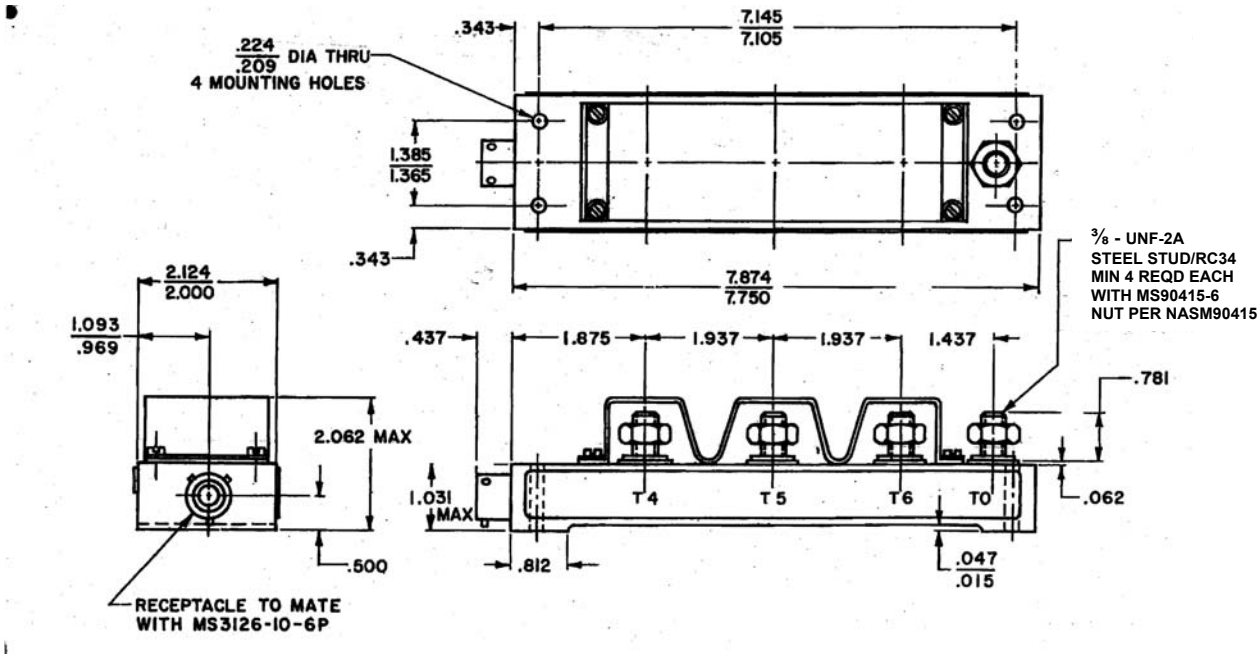


FIGURE 4. Ground end current transformer, weight 2.0 pounds max, P/N M21480/12-6.

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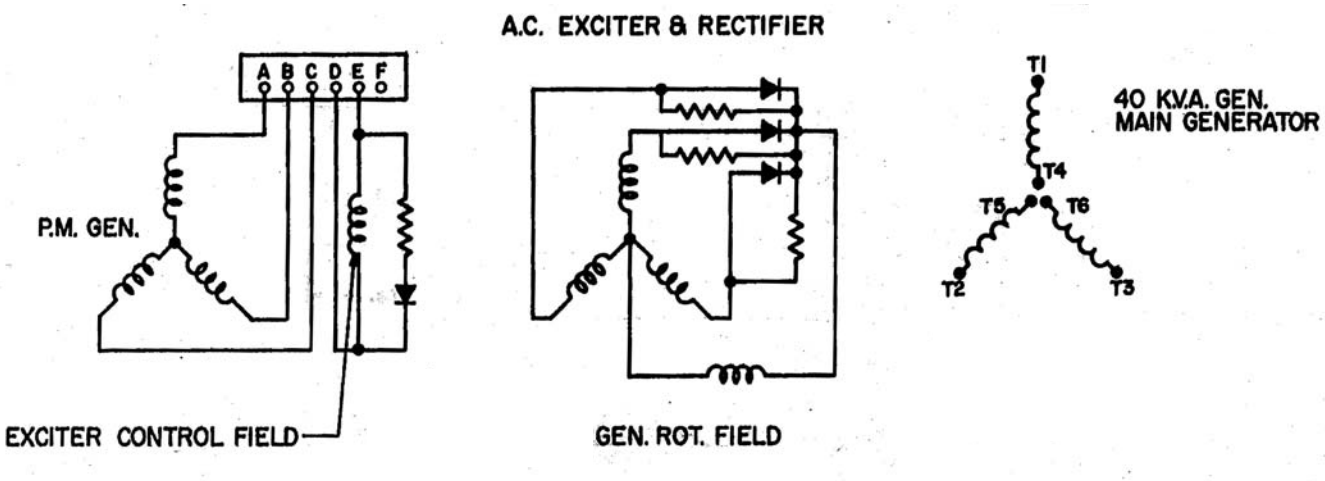


FIGURE 5. Generator circuit diagram.

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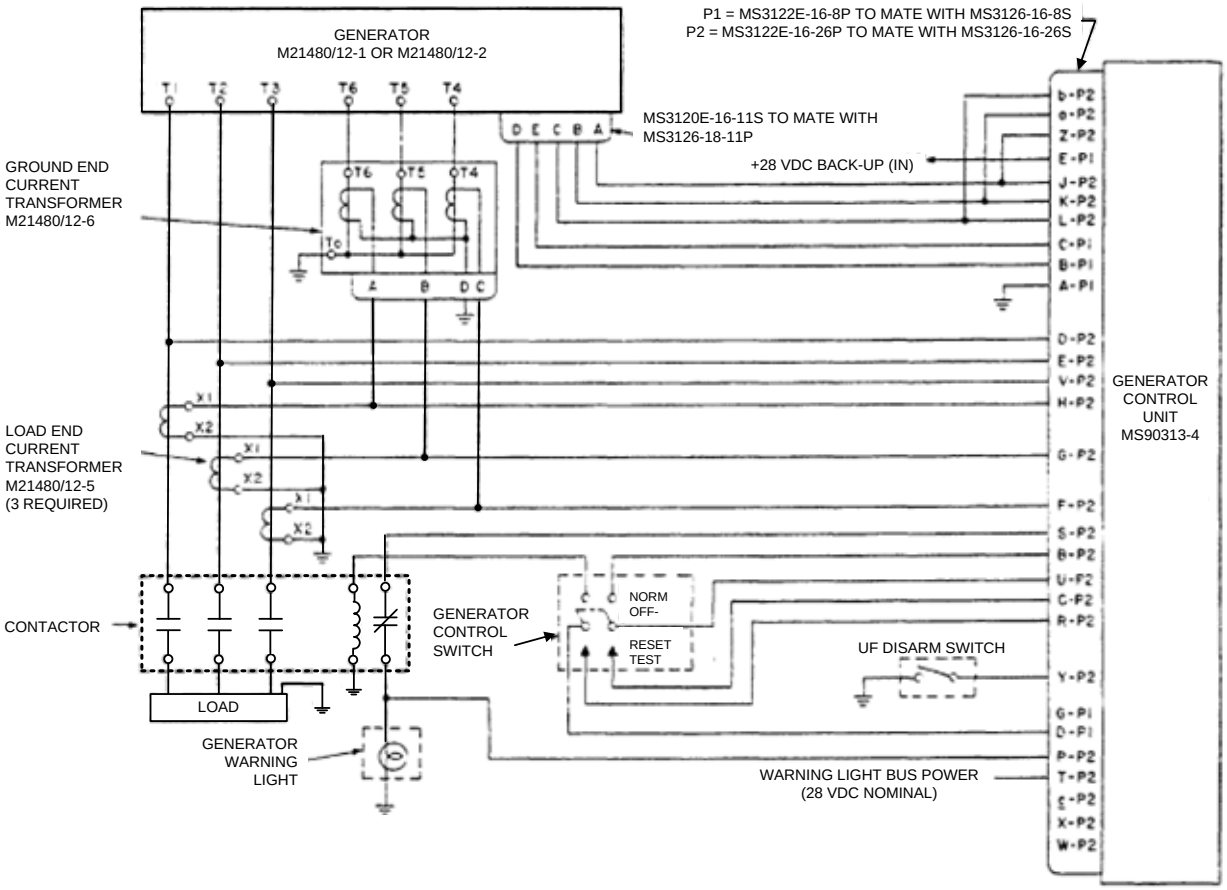


FIGURE 6. Test circuit.

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ALTITUDE (Feet)	INLET AIR TEMPERATURE DEGREES CENTRIGRADE (C°)	GENERATOR MAXIMUM CONTINUOUS LOAD	
		.75 POWER FACTOR	UNITY POWER FACTOR
0 - Sea Level	10	60	63
0 - Sea Level	20	61	67
0 - Sea Level	10	66	68
0 - Sea Level	0	68	71
0 - Sea Level	-20	72	75
10,000	20	55	57
10,000	10	58	60
10,000	0	60	62
10,000	-20	63	68
15,000	20	50	51
15,000	10	51	52
15,000	0	55	57
15,000	-20	59	61
25,000	20	42	42
25,000	10	43	45
25,000	0	47	49
25,000	-20	52	54

FIGURE 7. Generator altitude rating chart.

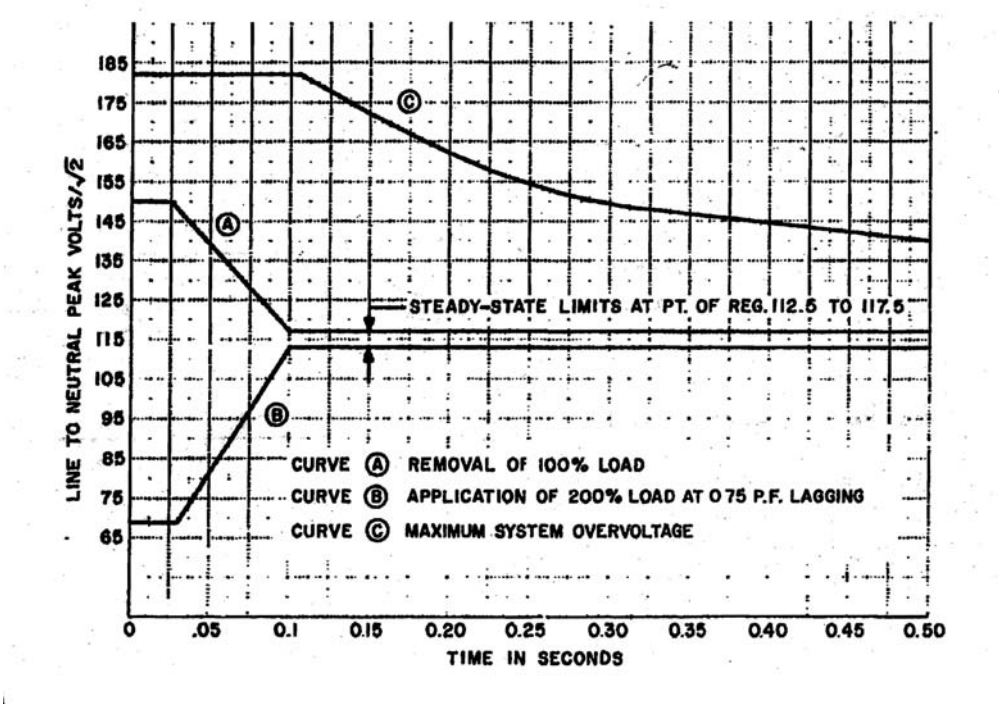


FIGURE 8. Transient, steady-state and overvoltage limits.

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REQUIREMENTS:

The requirements of MIL-PRF-21480 apply with the following exceptions:

Delete the following paragraphs:

3.4.2.2 through 3.4.2.8, 4.5.20, and 4.5.36.

3.4.2 Delete and add: Environmental requirements. The system shall meet the requirements of MIL-E-81910 unless otherwise specified by this specification sheet.

3.4.2.1 Delete and add: Temperature and altitude. The generator shall meet the temperature-altitude requirements of MIL-E-81910 for blast cooled equipment. When operated under self-cooled conditions, the generator shall meet the requirements of figure 7 of the specification sheet. The control components shall operate continuously when subjected to the temperature-altitude requirements of MIL-E-81910 for self-cooled equipment.

3.4.8 Add the following sentence at the end of the paragraph: "The voltage limits on figure 8 of this specification shall apply."

3.4.9.3 Add: The generator system shall be designed so that upon removal of any connector plug other than those of the current transformers, the system shall be deenergized.

Add 3.4.9.3.1 Wiring schematic. Generator wiring shall be in accordance with figure 5.

3.4.9.6 Add: The generator shall be self-cooled and in addition shall be provided with means for installation of an air inlet tube.

3.4.10 Change "5000 hours" to "4000 hours."

3.4.11 Change "Systems components" to "Systems components except current transformers."

3.5.1. Add the following sentence to the end of the paragraph: "The transient and steady state voltage shall remain within the limits on figure 8 of this specification sheet for all balanced load conditions while the system is operating under all the environmental and load conditions.

3.5.7.2 Delete "figure 13 for class N and figure 14 for class W," and replace with, "figure 8 of the specification sheet."

3.5.7.6.b Delete, "between 15 and 25 amperes," and replace with, "18 ±3 amperes," in the last sentence of the paragraph.

3.10 Delete and add: Reliability. The system MTBF (Mean Time Between Failures) shall be not less than 1200 hours. This figure shall apply to failures which will cause system operation to deviate from performance requirement.

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4.2.2.b In the last sentence of the paragraph, delete, “50 hours,” and replace with “100 hours.”

4.5.12 Delete and add: Electromagnetic interference. The system shall be subjected to an electromagnetic interference test in accordance with MIL-E-81910 for Class IIIB equipment.

4.5.13 Delete and add: Wave form. The crest factor and harmonic content, line-line and line-neutral, of the output voltage shall be determined under the following conditions:

a. At generator speed of 6000 RPM for no load and in 20 percent load increments to full rated load, unity power factor.

b. Condition (a) shall be repeated for 0.75 power factor.

c. Conditions (a) and (b) shall be repeated at generator speeds of 5700 and 6300 RPM.

The following requirements must be met during all above tests. The line-neutral crest factor shall be 1.41 ±10 percent. No harmonic shall exceed 3 percent of the fundamental.

4.5.20.1 Delete and add: Salt fog, control components. Control components shall be subjected to a salt fog test in accordance with MIL-E-81910, except the duration of the test shall not exceed 25 hours. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of rated and 200% load, and protective functions.

4.5.20.2 Delete and add: Oil-salt water ingestion, generators. The generator shall be subjected to an oil-salt water ingestion test in accordance with MIL-E-81910. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200 percent load.

4.5.21 Delete and add: Fungus resistance. System components shall be subjected to a fungus test in accordance with MIL-E-81910, with the installation instructions and mounting nuts attached. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200 percent load, and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible.

4.5.22 Delete and add: Sand and dust. The generator shall be subjected to a sand and dust test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its building characteristics and the effect of application and removal of rated and 200 percent load.

4.5.24 Delete and add: Shock. System components shall be subjected to a shock test in accordance with MIL-E-81910. At the completion of procedure I, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of

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rated and 200 percent load, and protective functions. There shall be no failure of the mounting attachments and the test item shall remain in place and not create a hazard during procedure III.

4.5.25 Delete and add: Humidity. System components, except those tested under 4.5.20.2, shall be subjected to a humidity test in accordance with MIL-E-81910. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200 percent load, and protection functions.

4.5.35 Delete the paragraph and add: All systems components except the generator shall be subjected to the acceleration tests of MIL-E-81910. At the completion of the tests the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200 percent load, and protective functions.

NOTES

1. Components enclosed in dotted lines are shown for interconnections only and are not furnished under this specification sheet.

2. Dimensions in inches. Unless otherwise specified, tolerances: linear ±.031, angular ±1°.

3. All outline dimensions are limiting dimensions only.

4. Thread dimensions and designations shall be interpreted in accordance with FED-STD-H28 and ASME Y14.6, respectively.

5. The difference in the M21480/12-1 and M21480/12-2 generators is that the M21480/12-1 uses Vanadium Permendur steel.

ITEM	PART NUMBER
Complete System (Consists of one Generator, one Generator Control Unit, three Load End Current Transformers, and one Ground End Current Transformer	M21480/12-40
Generator	M21480/12-1
Generator	M21480/12-2
Generator Control Unit	M90313-4
Load End Current Transformer	M21480/12-5
Ground End Current Transformer	M21480/12-6

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-21480, this document references the following:

ASME Y14.6	FED-STD-H28	MS3110	MS3120	MS3122
MS3126	SAE-AS5131	MS14169	MS18056	MIL-E-81910
MS90313	NASM90415			

Custodian:
Navy - AS

Preparing activity:
Navy - AS
(Project 6115-2012-002)

Review activity:
DLA - GS

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